# Supplementary Material: Auditing Metrics for Synthetic Data Utility A Cross-Domain Study of Appearance and Geometry

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## A. Implementation Details

Unless otherwise stated, the joint-summary analyses use DINOv3 for appearance and MASt3R for geometry. Appendix B reports the exhaustive sweep over all 32 appearance–geometry pairs; the leading configurations lie within $0.003$ of one another. We use the same pair throughout for consistency, while the full grid is part of the broader metric-family evaluation. For appearance, we use a DINOv3 ViT-Large checkpoint. Each image is resized to $518 \times 518$, normalized with ImageNet mean and standard deviation, and encoded into patch tokens. A single global representation is obtained by average-pooling the normalized patch-token grid. Appearance similarity is computed as cosine similarity between the resulting $\ell_2$-normalized embeddings.

For geometry, we use the MASt3R ViT-Large model. Each real–synthetic image pair is processed at resolution $512$, and dense descriptor maps are extracted with MASt3R. Descriptor maps larger than $256 \times 256$ are bilinearly downsampled before matching. Correspondences are obtained using mutual nearest-neighbor matching in descriptor space. A fundamental matrix is estimated with `cv2.findFundamentalMat` using `USAC_MAGSAC` with reprojection threshold $3.0$, confidence $0.99$, and up to $1000$ iterations. The number of inlier correspondences returned by this procedure defines $G(r_i, s_j)$. If fewer than 8 matches are available, or geometric verification fails, the inlier count is set to zero. This raw count is not normalized by the number of tentative matches, visible area, or texture density. Across unrelated domains it can therefore reflect matcher opportunity as well as geometric compatibility; we treat it as a correspondence signal rather than a domain-invariant measure of geometry.

In the released implementation, the main SADGE configuration is fit on the full benchmark collection of 21 dataset variants. The final released configuration uses the constrained bilinear form with parameters $a = 0.2818$, $b = 0.2659$, $c = 0.4524$, and normalization statistics $\mu_G = 7.0248$, $\sigma_G = 2.0445$, $\mu_A = 0.6242$, $\sigma_A = 0.2163$. Because Pearson correlation is invariant to a positive rescaling of the score, $(a, b, c)$ is identified only up to a positive scale factor; we therefore report the triple at the scale $a + b + c = 1$, so that $c$ is directly the normalized weight of the interaction term, $c/(a + b + c) = 0.452$. For efficiency, pair-level metric computations are cached and reused across runs. The runtime bottleneck for evaluating SADGE on a new synthetic–real dataset pair is correspondence estimation with MASt3R. The sensitivity analysis in Fig. 2 and Appendix C shows that the qualitative conclusion does not depend on the exact calibrated coefficient triple.

Table 1. Candidate-pool-size sensitivity on the 15 variants with caches at $k \in \{1, 3, 10\}$. Coefficients are held fixed. This tests pool size on one cached draw, not variation across random seeds.

| $k$ | 1 | 3 | 10 |
|---|---|---|---|
| $r$ | 0.710 | 0.856 | 0.878 |

The pool-size values rise from $k = 1$ to $k = 3$ and change less from $k = 3$ to $k = 10$ on this subset. This pool-size analysis uses one cached candidate draw and does not establish robustness to resampling. We assess resampling separately on six polyp variants using five candidate pools.

## B. Ablation of Similarity Metric Combinations

The main study finds that standalone appearance and geometry proxies respond differently as the benchmark composition changes. This appendix asks whether the observed benefit of combining those signal families depends on one particular encoder or matcher. To determine the configuration for SADGE, we conducted an evaluation of various foundation models and standard metrics. Specifically, we computed the SADGE score using four different geometry-based methods (SSIM, SuperPoint, MASt3R, LoFTR) crossed with eight appearance-based approaches (FID, DINOv2, DINOv3, SigLIP, SAM3, PSNR, CLIP, LPIPS). We evaluated each of these 32 configurations

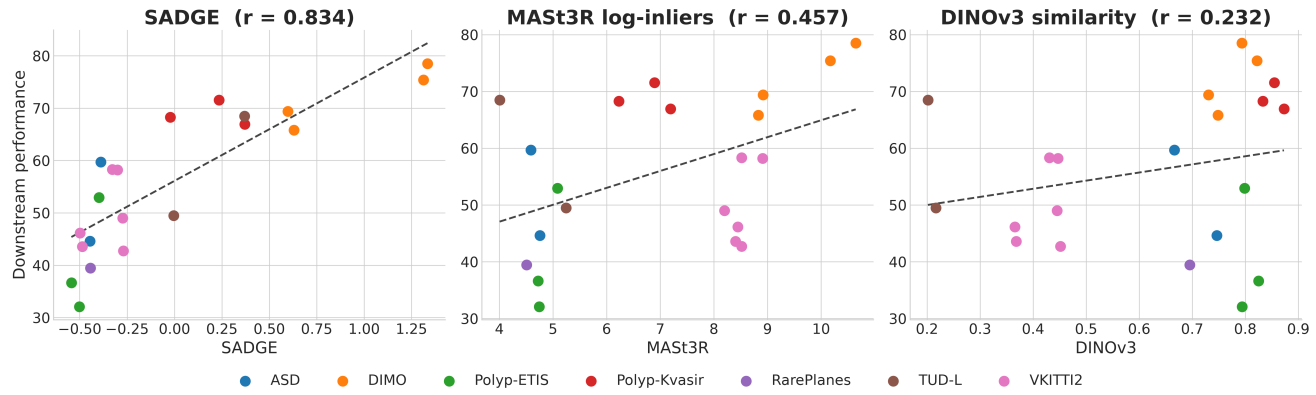



Figure 1. Downstream score versus the exploratory SADGE summary, log MASt3R inlier count, and DINOv3 similarity over the 21 dataset variants, colored by benchmark family. The apparent pooled alignment includes substantial between-family structure and must not be interpreted as held-out prediction.

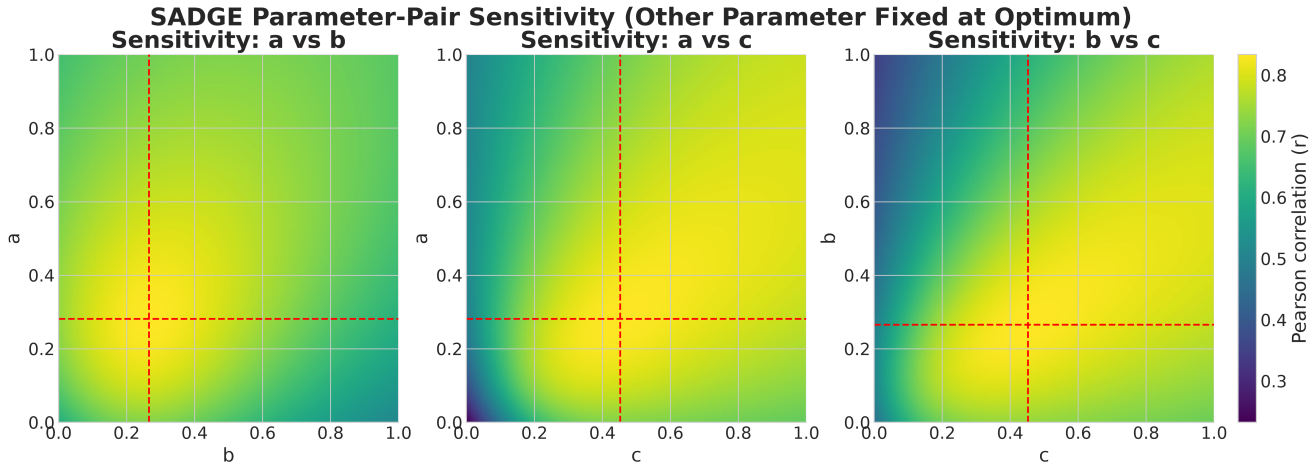



Figure 2. Descriptive sensitivity of the in-sample SADGE correlation to coefficient pairs $(a, b)$, $(a, c)$, and $(b, c)$, with the third coefficient fixed. This is a fit-sensitivity visualization, not held-out validation.

across all seven public or author-permitted benchmark families, encompassing 21 dataset-level variants and 87,880 image pairs. MASt3R $\times$ SigLIP leads the sweep at $r = 0.8366$ and MASt3R $\times$ DINOv3 is third at $r = 0.8337$, with the three leading configurations spread over only $0.003$. A $0.003$ gap over 21 observations does not identify one encoder over another: a paired bootstrap over variants puts the SigLIP-minus-DINOv3 difference at $+0.003$ with a $95\%$ interval of $[-0.055, +0.069]$. We therefore use MASt3R $\times$ DINOv3 consistently rather than declaring a unique winner from the grid. The more important family-level pattern is that the eight highest-scoring configurations all use MASt3R or LoFTR, whose row means ($0.749$ and $0.699$) exceed those of SuperPoint+LightGlue ($0.600$) and SSIM ($0.439$). The evidence therefore supports a contribution from dense structural correspondence, while the exact appearance encoder is comparatively interchangeable.

## C. Alternative Joint Summary Functions

We searched for the SADGE fusion form by enumerating sixteen candidate equations that combine the (z-scored) geometry score $g$ and appearance score $a$ into a single scalar. Each equation's free parameters were fit to maximize the Pearson correlation between SADGE and the downstream reference on the full pool of $n = 21$ (dataset, variant) rows, using the canonical metric pair MASt3R-inliers $\times$ DINOv3 similarity. To avoid local optima, one-parameter forms are optimized on a dense grid, two-parameter forms with 200 random multi-starts of L-BFGS-B within each parameter bound, and the polynomial forms by an exhaustive search of the coefficient simplex followed by a local refinement. Table 3 reports $|r|$ for each equation, sorted high-to-low. The constrained interaction polynomial has the largest descriptive correlation and ties its unconstrained counterpart because the latter's optimum lies on the positive face of the parameter cube. The margin over the alternatives is sub-

Table 2. Pearson correlation ($r$) of SADGE scores across different combinations of geometry-based and appearance-based similarity metrics, over all 21 dataset variants. The best performing configuration (MASt3R × SigLIP) is highlighted in bold; the configuration used throughout the paper (MASt3R × DINOv3) is underlined.

| **Geo ↓ \ App →** | **FID** | **DINOv2** | **DINOv3** | **SigLIP** | **SAM3** | **PSNR** | **CLIP** | **LPIPS** |
|---|---|---|---|---|---|---|---|---|
| SSIM | 0.4637 | 0.4753 | 0.3405 | 0.3228 | 0.3528 | 0.6036 | 0.3210 | 0.6315 |
| SP+LG | 0.3565 | 0.6362 | 0.6565 | 0.6370 | 0.5261 | 0.7245 | 0.6733 | 0.5909 |
| MASt3R | 0.5823 | 0.8040 | <u>0.8337</u> | **0.8366** | 0.7582 | 0.6972 | 0.8350 | 0.6407 |
| LoFTR | 0.5850 | 0.7576 | 0.7508 | 0.7326 | 0.6669 | 0.7029 | 0.7273 | 0.6686 |

stantial on this pool: the gap to the next family is $0.118$, and no other form lands within $0.02$ of the highest value. The forms that come closest are the ones that also contain a multiplicative term between the two signals, while the purely additive and blending forms trail by more than $0.25$. The equation-level reading is therefore consistent with the study's broader claim: what matters is that the summary lets one signal modulate the other, and the constrained polynomial is the simplest and most directly interpretable way of writing that.

## D. Canonical Variant Table

Table 4 prints the table; Table 7 summarizes the families. Tables 6 and 5 document the full protocol behind every row. Table 6 is per variant: the real target dataset, the synthetic source, the downstream score, the metric exactly as the source names it, the direction of transfer, the source publication, whether the correspondence is genuinely aligned or retrieval-based, the number of real queries, and the candidate-pool size with its seed. Table 5 is per family, because the downstream model and real evaluation split are read from one source comparison. Within each source benchmark we take the reported scores from one table of one publication, but this does not guarantee that every source controls the number of training images or all source-specific construction details. In particular, TUD-L compares approximately 350K PBR images with 50K render-and-paste images. Its downstream difference therefore combines rendering paradigm and training-budget effects and must not be interpreted as an isolated effect of synthetic image compatibility. Every field in Table 5 is quoted or closely paraphrased from that publication; where a source does not state a field we write *not stated* rather than inferring it.

Two entries deserve emphasis because they are easy to read past. Virtual KITTI 2 is the reverse-transfer family discussed above, and its model is not trained by the source at all: it is AdapNet++ under the authors' released CityScapes weights, evaluated on the synthetic variants. And the ASD score is the *crop-class* IoU of a table that reports weed and crop separately, not an overall mIoU; the two ASD variants are that source's plain synthetic and domain-adapted datasets, both trained at a 12k:0 synthetic-to-real ratio and both evaluated on the same 3,981-image real soybean hold-out.

The benchmark is released as a single machine-readable table with one row per variant, carrying the family, variant, real target, synthetic source, downstream task and metric, raw downstream score, transfer direction, source reference, pairing mode, number of real queries, candidate-pool size, candidate seed, every standalone proxy value, both SADGE components before and after normalization, and the resulting joint scores. Every number in this paper is derived from that table by the released scripts.

Two properties of the protocol are worth stating explicitly because they are easy to get wrong. First, the candidate set is drawn once per real query and reused by every pairwise metric, so all retrieval-based metrics see identical candidates and differ only in which candidate they select; the draw is keyed on (seed, variant, image) so it is independent of run order and of how variants are split across jobs. Second, FID is computed once per variant at the distribution level and never enters the per-query retrieval step, and LPIPS is minimized rather than maximized during selection. The candidate seed is fixed at $42$ throughout.

Sensitivity to the candidate draw is quantified in App. G, on the two families whose source imagery is still available to us: the pair-level metrics were recomputed under four further pools there, and the resulting 25 benchmark tables move no pooled correlation by more than $0.026$. The remaining eight retrieval variants cannot be re-drawn, so the effect is bounded on part of the benchmark and not on the rest. All reported analyses and manuscript quantities are reproducible from the released cached measurements. Full recomputation from raw pixels has narrower coverage: it requires access to the original datasets and gated checkpoints, ASD/SynSoy is permission-based, and the archived sources do not currently permit independent candidate-pool redraws for eight retrieval variants. We therefore distinguish reproducibility of the reported statistical analysis from bit-level regeneration of every raw metric measurement. The synthetic pools for five of the seven families are larger than the candidate budget, so the unmeasured draw effect cannot be assumed irrelevant.

Table 3. Sixteen candidate fusion equations evaluated on the canonical SADGE pair (MASt3R-inliers, DINOv3) over $n = 21$ (dataset, variant) rows. Parameters were fit to maximize Pearson correlation against the downstream reference. The interaction polynomial form is selected for SADGE; the constrained ($\alpha, \beta, \gamma \geq 0$) and unconstrained variants tie because the unconstrained optimum lies on the positive face of the parameter cube. No other form lies within 0.02 of the best, and the nearest alternatives are the other forms with a multiplicative interaction.

| # | Equation | Fitted params | $\lvert r \rvert$ |
|---|---|---|---|
| 1 | `constrained_polynomial` | (0.282, 0.266, 0.452) | **0.834** |
| 1 | `interaction_polynomial` | (0.282, 0.266, 0.452) | 0.834 |
| 3 | `generalized_mean_softplus` | (0.486, −2.000) | 0.716 |
| 4 | `tversky_softplus` | (2.000, 2.000) | 0.705 |
| 5 | `eccv_synergistic_gating` | ($\tau = 1.862,$) | 0.697 |
| 5 | `fbeta_softplus` | ($\beta = 0.961,$) | 0.697 |
| 7 | `generalized_mean` | (0.274, 0.060) | 0.655 |
| 8 | `tversky_index` | (1.317, 0.129) | 0.653 |
| 9 | `fbeta_score` | ($\beta = 3.228,$) | 0.652 |
| 9 | `weighted_harmonic` | (0.088,) | 0.652 |
| 11 | `softplus_linear` | (0.587,) | 0.573 |
| 12 | `atan_blend` | (0.584,) | 0.526 |
| 13 | `robust_saturating_sum` | (0.570,) | 0.517 |
| 14 | `sadge_linear` | (0.661,) | 0.514 |
| 15 | `logsumexp_blend` | (0.678, $\tau = 0.100$) | 0.502 |
| 16 | `gated_blend` | ($\tau = 0.100,$) | 0.459 |

Table 4. The 21 benchmark variants with downstream score $y$ (AP, mIoU/IoU or AR as published) and every standalone proxy value as used in the paper (raw orientation; LPIPS and FID are distances, correspondence columns are mean inlier counts). Generated from `tables/benchmark_variants.csv`, which additionally carries target provenance, transfer direction, pairing mode, query counts and the candidate seed.

| Family | Variant | Task | $y$ | SSIM | PSNR | LPIPS | FID | CLIP | SigLIP | SAM3 | DINOv2 | DINOv3 | MASt3R | SP+LG | LoFTR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ASD | cotton | seg | 59.70 | 0.044 | 12.091 | 0.542 | 146.7 | 0.837 | 0.803 | 0.910 | 0.776 | 0.667 | 97.431 | 23.640 | 24.063 |
| ASD | domain_adapted | seg | 44.61 | 0.050 | 12.047 | 0.521 | 69.381 | 0.898 | 0.903 | 0.960 | 0.833 | 0.746 | 115.4 | 25.781 | 24.664 |
| DIMO | randpose_randlight | det | 69.40 | 0.297 | 15.438 | 0.585 | 88.635 | 0.826 | 0.874 | 0.976 | 0.800 | 0.731 | 7463.2 | 20.323 | 129.6 |
| DIMO | randpose_reallight | det | 65.80 | 0.278 | 14.801 | 0.576 | 99.646 | 0.832 | 0.874 | 0.976 | 0.809 | 0.748 | 6840.8 | 18.834 | 130.3 |
| DIMO | realpose_randlight | det | 75.40 | 0.351 | 12.189 | 0.394 | 70.356 | 0.866 | 0.907 | 0.976 | 0.919 | 0.822 | 26086.0 | 168.3 | 174.5 |
| DIMO | realpose_reallight | det | 78.50 | 0.528 | 14.815 | 0.212 | 176.3 | 0.805 | 0.834 | 0.979 | 0.928 | 0.794 | 41988.0 | 244.6 | 118.2 |
| Polyp-ETIS | ldm | polyp segmentation | 36.64 | 0.471 | 11.466 | 0.635 | 143.6 | 0.916 | 0.923 | 0.972 | 0.849 | 0.825 | 111.3 | 12.372 | 33.847 |
| Polyp-ETIS | polyp-ddpm | polyp segmentation | 52.95 | 0.535 | 11.942 | 0.675 | 197.8 | 0.888 | 0.856 | 0.967 | 0.845 | 0.799 | 160.7 | 14.959 | 30.296 |
| Polyp-ETIS | singan-seg | polyp segmentation | 32.09 | 0.555 | 12.361 | 0.657 | 146.9 | 0.905 | 0.897 | 0.975 | 0.824 | 0.794 | 113.8 | 13.923 | 28.673 |
| Polyp-Kvasir | ldm | polyp segmentation | 66.94 | 0.617 | 13.206 | 0.563 | 136.1 | 0.917 | 0.944 | 0.971 | 0.878 | 0.873 | 1326.3 | 22.400 | 55.330 |
| Polyp-Kvasir | polyp-ddpm | polyp segmentation | 71.56 | 0.695 | 13.694 | 0.583 | 124.2 | 0.912 | 0.891 | 0.975 | 0.885 | 0.855 | 984.7 | 27.760 | 49.170 |
| Polyp-Kvasir | singan-seg | polyp segmentation | 68.28 | 0.702 | 13.840 | 0.574 | 130.9 | 0.906 | 0.923 | 0.975 | 0.851 | 0.833 | 507.0 | 20.720 | 40.170 |
| RarePlanes | test | det | 39.46 | 0.114 | 13.934 | 0.694 | 118.0 | 0.770 | 0.815 | 0.893 | 0.831 | 0.695 | 89.628 | 13.341 | 20.487 |
| TUD-L | train_bad_render_backgrounds | pose | 49.50 | 0.266 | 10.781 | 0.789 | 304.7 | 0.600 | 0.615 | 0.872 | 0.303 | 0.217 | 188.7 | 23.745 | 33.942 |
| TUD-L | train_good_render | pose | 68.50 | 0.169 | 11.259 | 0.756 | 342.6 | 0.644 | 0.638 | 0.873 | 0.312 | 0.202 | 53.915 | 18.243 | 26.637 |
| VKITTI | clone | seg | 58.22 | 0.398 | 9.861 | 0.654 | 122.6 | 0.709 | 0.802 | 0.950 | 0.677 | 0.447 | 7383.3 | 105.5 | 119.9 |
| VKITTI | fog | seg | 43.59 | 0.101 | 10.474 | 0.749 | 171.7 | 0.665 | 0.762 | 0.922 | 0.610 | 0.368 | 4456.5 | 29.211 | 26.739 |
| VKITTI | morning | seg | 42.73 | 0.303 | 9.016 | 0.689 | 130.9 | 0.696 | 0.787 | 0.954 | 0.660 | 0.451 | 5004.0 | 79.209 | 86.300 |
| VKITTI | overcast | seg | 58.31 | 0.290 | 9.402 | 0.685 | 121.2 | 0.684 | 0.794 | 0.936 | 0.656 | 0.431 | 5010.0 | 81.036 | 86.878 |
| VKITTI | rain | seg | 46.14 | 0.194 | 9.425 | 0.680 | 118.4 | 0.677 | 0.759 | 0.915 | 0.636 | 0.366 | 4649.1 | 81.826 | 71.049 |
| VKITTI | sunset | seg | 49.01 | 0.245 | 8.560 | 0.678 | 129.8 | 0.687 | 0.779 | 0.964 | 0.659 | 0.445 | 3625.5 | 77.341 | 76.275 |

## E. Transfer Direction and the Family Centering Convention

This section clarifies transfer direction and compares the reported family-mean decomposition with an alternative based on family medians.

**Virtual KITTI 2 is evaluated in the reverse direction.** Six of the 21 rows come from Virtual KITTI 2, and they are the only rows where the synthetic data is the *test* set rather than the training set. The source [1] reports, in its `ss_rgb` table, a semantic-segmentation model trained on real imagery (Cityscapes) and evaluated on each synthetic variant, and it names the reported measure *mAP*, not mIoU; our score is the mean over scenes 01, 02, 06, 18 and 20. This is the reverse-transfer case that the main paper's protocol admits explicitly, and the released variant table records it in the `transfer_direction` column. Earlier descriptions of this family as "mIoU-style" denote the quantity the source reports as mAP, as now given in Tables 6 and 7.

Because the direction differs, we report the benchmark restricted to the 15 strictly synthetic-to-real rows (six fami-

lies; Virtual KITTI 2 removed) in Table 8. The released calibration, frozen, reaches $r = 0.841$ on that subset against $0.834$ pooled, with $\rho = 0.875$; refitting the coefficients and normalization on the subset gives $r = 0.857$. The strongest standalone proxy on the same rows is LoFTR at $r = 0.633$. The conclusion is therefore not carried by the reverse-transfer family: removing it leaves the joint score ahead of every standalone proxy by a margin comparable to the full pool. This check does not make the remaining targets commensurate and does not remove the TUD-L training-budget confound; it isolates transfer direction only.

**The reported decomposition uses family means.** The values in Table 2 of the main paper were computed by subtracting each row's family mean for the within-family component and by correlating the seven family means for the between-family component. Table 9 repeats the analysis using family medians. SADGE remains the leading score under both: its within-family correlation changes from $0.524$ to $0.509$, and its between-family correlation from $0.893$ to $0.904$. The ordering of all four scores is unchanged. The mean-based values reproduce Table 2.

## F. Heterogeneous Downstream Scales

The downstream targets are not measurements on a common scale. They combine COCO AP, source-defined mAP, crop-class IoU, polyp IoU, and BOP average recall under seven different models and protocols. A pooled Pearson correlation over these values compares unlike quantities and is not invariant to family-specific changes of units. It therefore has no interpretation as correlation with one universal downstream variable; we use it only as a descriptive summary of the assembled benchmark. Table 10 reports five family-aware readings and a strictly synthetic-to-real subset alongside that summary. These sensitivity criteria, rather than the raw pooled value alone, test whether the qualitative ordering survives removal of cross-family location, scale, weighting, or transfer-direction effects.

The first row is the raw pooled correlation. The next three remove each family's location: centring every row by its family mean, by its family median, and $z$-standardising score and target inside each family before pooling, which removes the family's spread as well as its location. The fifth weights each family equally instead of by its variant count, so that Virtual KITTI 2's six rows do not outvote RarePlanes' one. The last restricts the pool to strictly synthetic-to-real transfer. Families contributing one variant carry no within-family information and drop out of rows two to five, leaving 20 rows over six families.

Two things follow.

First, the ranking is stable under every reported criterion that keeps a magnitude. These family-aware and strictly synthetic-to-real readings are the primary robustness interpretation of the benchmark; the raw pooled value is retained only as its descriptive starting point. The joint score leads on the raw pooled correlation, under both centrings, under family-wise $z$-standardisation, on the family-balanced macro average, and on the synthetic-to-real subset. The margin narrows as more of the family scale is removed – from $+0.834$ pooled to $+0.509$ under median centring – showing that part of the pooled alignment is between-family structure. These are still low-sample summaries of a score selected on the assembled benchmark, not independent evidence of universal calibration.

Second, the family-balanced macro average is the criterion on which the standalone proxies fail outright: LPIPS averages $-0.021$ and log LoFTR $-0.234$ across the six families, because each is strong on some families and inverted on others, whereas the joint score averages $+0.680$. This is the same instability the main paper reports from the family-exclusion panels, measured as one number. It should be read with its degrees of freedom in view: two of the six families contribute two variants each, so their per-family correlation is $\pm 1$ by construction, and the macro average over six families is a coarse statistic. RarePlanes contributes no within-family comparison. The study therefore supports benchmark-dependent metric behavior, but not a causal rule assigning an optimal metric family from a domain-gap label.

## G. Sensitivity to the Retrieval Pool Draw

Fourteen of the 21 variants are scored in retrieval mode: each real query is paired with the most similar of $k = 10$ synthetic candidates, and those candidates are drawn once, under candidate seed 42. The draw is a design choice the benchmark has not paid for. A different draw offers different candidates, so the retrieved pair differs, so every proxy value on that row differs, so the fit differs. This section measures how large that effect is.

**What can and cannot be re-drawn.** The measurement is possible on two of the seven families. The polyp imagery is the only retrieval source still available to us, and four further candidate pools were drawn for it – seeds 7, 13, 21 and 34 – with the metric stack re-run in full on each. With the canonical seed that gives five independent pools for six of the 21 variants, covering nine of the twelve proxies; CLIP, SigLIP and SAM3 were not re-run and hold their canonical values throughout. The remaining eight retrieval variants (ASD, DIMO, RarePlanes, TUD-L) cannot be re-drawn, because their source imagery is no longer available to us; this is a limitation of our own archive, not of the protocol. The seven aligned variants have no candidate pool and are unaffected by construction. The seed-42 values recovered from the re-run agree with the canonical variant table to within

$10^{-6}$ on every proxy, which is the check that the re-runs are the same computation.

**At the level of a single proxy value, the draw matters.** Table 11 puts each proxy's seed standard deviation next to the standard deviation across the three variants of the same family – the signal it is being asked to resolve. The appearance proxies are stable: DINOv3 moves by 0.13 of the between-variant spread on Kvasir, LPIPS by 0.26, FID by 0.10. The correspondence proxies are not. MASt3R's seed spread is 0.56 of the between-variant spread on Kvasir and 0.40 on ETIS; LoFTR's is 0.39 on Kvasir. In raw terms the MASt3R inlier count for Kvasir `singan-seg` runs from 507 to 1281 across the five pools, a factor of 2.5, on a family whose three variants span roughly the same range. The consequence is direct: on six of the eighteen proxy-family combinations the pool draw changes the order of the variants inside the family, and four of the six involve a correspondence proxy. Correspondence counts are the most pool-sensitive measurements in the benchmark, which is unsurprising – retrieval selects the candidate a proxy scores highest, and inlier counts have the heaviest right tail of any proxy here.

**The measured pooled effect is small on the available rows.** Crossing the five Kvasir pools with the five ETIS pools gives 25 benchmark tables differing only in the retrieval draw on those six rows. Each is re-analysed end to end. Across all 25, the pooled correlation of every proxy moves by at most 0.026 (MASt3R; LoFTR 0.020, LPIPS 0.005), and *the induced ranking of the nine redrawn proxies is identical in all 25 tables* – LPIPS first in every one, LoFTR second, MASt3R fourth. The refitted SADGE correlation ranges over $[0.828, 0.853]$ against a canonical 0.834; the normalised coefficients move by at most 0.008, with the interaction share $c/(a{+}b{+}c)$ confined to $[0.451, 0.454]$. Re-running the whole 112-split family-disjoint protocol inside each of the 25 tables moves the median over $[0.714, 0.762]$, the negative fraction over $[0.179, 0.196]$ and the worst split over $[-0.744, -0.728]$; every claim in the released tables holds under every draw.

**Reading the two together.** Six rows of 21 are re-drawn, so a per-row perturbation of the size seen above is diluted by a factor of roughly $\sqrt{21/6}$ before it reaches a pooled statistic, and the perturbations on Kvasir and ETIS are independent. The conclusion is correspondingly narrow: the available redraws do not materially change the pooled ordering, but they cover only two of the six retrieval families. The single pool draw remains a material uncertainty for within-family readings of correspondence proxies, and the within-family results in App. F do not propagate this variance. We have bounded the effect on two retrieval families and cannot bound it on the other four.

## H. Internal Validity of the Pooled Results

All pooled correlations in the paper are single point estimates over 21 dataset variants. This section reports how much they move under resampling and under the choices made in computing them, using the pooled Pearson criterion throughout; every table is generated from the canonical variant table by `scripts/internal_validity.py`.

**Bootstrap intervals.** Table 14 resamples the 21 variants with replacement (2,000 draws) and reports the 95% percentile interval of each pooled correlation and of the differences between scores. The SADGE point estimate of 0.834 has an interval of $[+0.731, +0.916]$, and its margin over LoFTR is $[+0.093, +0.626]$, which excludes zero. The margin between the strongest appearance-only and structure-only proxies (LoFTR minus LPIPS) is $[-0.363, +0.288]$ and does not: the benchmark separates the joint score from the standalone proxies, but not the two leading standalone families from each other.

**Influence of single variants.** Table 15 recomputes the pooled correlations with one variant removed. SADGE ranges over $0.807 - -0.855$ and LoFTR over $0.464 - -0.612$ across the 21 deletions, and in none of them does the joint score fall below the strongest standalone metric; the most influential rows are the DIMO pose variants. No single variant carries the pooled ordering.

**Transform of the correspondence counts.** Inside the joint score the inlier count enters as $\log(1 + \bar{G})$, whereas the standalone matchers in Fig. 2 of the main paper are correlated as raw mean counts. Table 16 gives both. On this pool the transform lowers the standalone matchers rather than raising them, MASt3R from 0.531 to 0.457 and LoFTR from 0.545 to 0.529, so the transform does not account for the margin between the joint score and the standalone bars in Fig. 2; against the log-transformed LoFTR the SADGE margin is $[+0.093, +0.658]$.

**Matched-capacity controls.** A fitted three-parameter score has two effective degrees of freedom that no standalone metric has. Table 17 fits the same non-negative bilinear form to two appearance metrics and to two geometry metrics. Appearance$\times$appearance pairs reach at most 0.641 and geometry$\times$geometry pairs at most 0.558, whereas geometry$\times$appearance pairs reach 0.834 (released) and 0.835. The capacity alone does not produce the pooled gain; the cross-kind pairing does, on this benchmark.

## I. Runtime and Coefficient Sensitivity

**Runtime profile.** Table 18 reports a runtime benchmark for metric computation on 1,000 real–synthetic pairs from

`dimo_small` on CUDA. For the SADGE configuration used in most reported results (DINOv3 and MASt3R), MASt3R is the dominant cost (271.46 s total, 3.68 pairs/s), and DINOv3 adds 31.52 s (31.73 pairs/s). The SADGE fusion step combines two dataset-level scores and adds negligible runtime compared with metric extraction. Additional baseline metrics are listed under the same benchmark setup for direct runtime comparison.

Runtime benchmarks were executed on an `x86_64` system with an NVIDIA GeForce RTX 3090 (driver 550.54.15, CUDA 12.4) and dual-socket AMD EPYC 7452 CPUs ($2\times32$ cores, 128 threads total, 1.5–2.35 GHz).

# J. Supplementary Evaluation Card for the Metric Study

This appendix documents the intended use, benchmark composition, asset provenance, license status, and reproducibility assumptions for the synthetic-data utility metric study. Following common practice in dataset and benchmark papers, we explicitly list the external datasets, pretrained models, and metric implementations used in our evaluation.

## J.1. Intended Use and Scope

**Purpose.** The primary artifact is a comparative evaluation of appearance- and structure-oriented metrics for synthetic-data utility. SADGE is the accompanying calibrated joint diagnostic: given a real target dataset and synthetic candidates, it summarizes the two signal families before training a model on the candidate dataset. Metric extraction and scoring are training-free *after calibration*, but constructing or adapting the calibration requires benchmark families with known downstream results. A practitioner without such results can apply the released calibration only as an exploratory reference and must validate the resulting choice in the target domain.

**Supported use.** The intended use is comparative dataset selection and diagnostic analysis of synthetic-to-real domain gaps in computer vision. SADGE is designed for settings where practitioners must choose among rendering configurations, domain-randomization schedules, synthetic variants, or generative data sources before expensive downstream training, provided that a relevant calibration has already been established and final real-domain validation remains available.

**Unsupported use.** SADGE should not be used as a sole deployment criterion for safety-critical systems, as a substitute for final real-domain validation, or as proof that a synthetic dataset is unbiased, fair, safe, or sufficient for a target application. SADGE is a proxy ranking metric, not a causal estimate of downstream performance.

**Monotonicity scope.** The bilinear interaction is not globally monotone because both inputs are signed z-scores: increasing one signal can lower SADGE when the other is sufficiently below the calibration mean. Both partial derivatives are positive simultaneously for 7 of the 21 observed variants. This behavior is part of the fitted exploratory interaction, not a general compatibility axiom, and the score should not be extrapolated outside the calibration support. A deployment-oriented successor should impose global monotonicity or validate the relevant operating region explicitly.

**Main claim supported by the benchmark.** Across seven public or author-permitted benchmark families and 21 dataset-level variants, standalone metric rankings change with benchmark composition, while appearance and geometry provide complementary evidence. A compact joint score has a higher descriptive correlation than every evaluated standalone proxy on the pooled benchmark and remains first in each family-exclusion sensitivity panel. The benchmark therefore supports the comparative signal-family finding under the assembled protocols; it does not identify a causal mapping from domain-gap mechanism to metric choice or establish universal calibration of the released coefficients.

## J.2. Dataset Licenses, Access Terms, and Attribution

We use existing public datasets or datasets for which the authors have permission. We do not redistribute third-party dataset images unless explicitly allowed by the corresponding license. Reproduction therefore requires obtaining each dataset under its original terms. We have checked the licenses listed below against the official dataset sources to the best of our knowledge.

**Dataset redistribution and levels of reproducibility.** The supplement does not redistribute raw third-party images. Users must obtain the underlying datasets from the original providers and comply with their licenses and terms of use. The released cached measurements, evaluation scripts, configuration files, and variant-level metadata reproduce every reported statistical analysis without raw imagery. Recomputing every metric from pixels is a stronger requirement and is not fully available: ASD/SynSoy requires permission, DINOv3 and SAM3 require acceptance of gated terms, eight retrieval variants cannot be independently re-drawn from the archived sources, and regenerated TUD-L render-and-paste imagery need not be bit-identical

to the cached variant. These constraints do not prevent verification of the reported analysis, but they limit independent regeneration of all raw measurements.

### J.3. Pretrained Models, Metric Implementations, and License Status

SADGE and the baselines rely on existing pretrained encoders, geometry matchers, and standard image-similarity metrics. Table 20 lists the main external model and metric assets. Exact package versions, checkpoint identifiers, and download instructions are required in the reproduction artifact.

**Use of non-commercial assets.** Some evaluated assets, including Virtual KITTI 2 and MASt3R, include non-commercial license terms. Our experiments are conducted for academic research. Any commercial reuse of the benchmark or released scripts must independently verify compatibility with all underlying dataset and model licenses.

### J.4. Known Limitations and Failure Modes

The evaluated proxies, including SADGE, may fail or become unreliable under the following conditions:

- the appearance encoder is insensitive to domain-specific artifacts relevant to the downstream task;
- the geometry matcher fails on textureless, transparent, reflective, repetitive, deformable, very small, or heavily occluded objects;
- real and synthetic images have little geometric overlap or very different camera viewpoints;
- downstream performance depends on labels, temporal cues, depth, multispectral channels, or task-specific annotations not visible in RGB;
- the synthetic-to-real gap is determined by annotation policy, label noise, class imbalance, or training-protocol effects rather than image similarity;
- target-domain images used for scoring are not representative of the downstream evaluation distribution;
- the candidate-pool draw changes rare high-scoring correspondences, especially in retrieval families for which repeated draws are unavailable;
- raw verified-inlier counts vary with tentative-match opportunity, visible area, texture density, and matcher behavior because they are not normalized by those quantities;
- calibration families use heterogeneous downstream scales or do not span the failure mechanism of an unseen family;
- a source benchmark changes training-set size or source-specific training details between variants, as in the TUD-L comparison;
- component or equation selection is performed on the same small benchmark used to assess the selected score;
- the target application is safety-critical and requires real-domain validation regardless of proxy metric ranking.

Overall, the study provides evidence that appearance and correspondence carry complementary information and that standalone proxy rankings are unstable under changes in benchmark composition. SADGE makes that relationship concrete through a transparent low-capacity probe. The pooled correlation is descriptive. The released artifact is therefore intended as an auditable basis for expanding synthetic-data utility benchmarks, not as a final or universally calibrated dataset-selection rule.

Table 5. Per-family downstream provenance. Each field is given as the source publication states it; *not stated* means the source does not report it. The downstream model and real evaluation split are shared within a family, but source-specific training budgets need not be: TUD-L uses approximately 350K PBR images versus 50K render-and-paste images.

| Family | Field | Value as stated by the source |
|---|---|---|
| DIMO | Downstream model | Mask R-CNN, ResNet-101 feature detector |
| | Training recipe | Transfer learning from COCO weights with the feature detector frozen, plus data augmentation (colour and translation); 100 epochs, SGD, lr 0.001, momentum 0.9, batch size 4, 1,000 images per epoch |
| | Training-set size | 1,775 images per variant; the source reduces every dataset 'to the size of the real dataset, which is 1775 images' |
| | Real evaluation split | Unseen real DIMO test images |
| | Metric | COCO AP (IoU@[0.5:0.95]) |
| | Source | Vanherle et al., Analysis of Training Object Detection Models with Synthetic Data, BMVC 2022 (arXiv:2211.16066), Fig. 3, 'Both' bars of the equal-size experiment |
| VKITTI | Downstream model | AdapNet++, the authors' released pretrained models |
| | Training recipe | Not trained for this experiment: 'We use the pre-trained models provided by the authors. These models were trained using the CityScapes dataset', on 11 CityScapes categories; scored on the 7 classes plus background common to both datasets |
| | Training-set size | not stated (CityScapes) |
| | Real evaluation split | Virtual KITTI 2 scenes 01, 02, 06, 18 and 20; our score is the mean over those five scenes. The source's real reference point is the 200 KITTI images with segmentation ground truth (mAP 58.07) |
| | Metric | mAP, as the source names it |
| | Source | Cabon et al., Virtual KITTI 2 (arXiv:2001.10773), Tab. 14 'RGB semantic segmentation mAP results' |
| TUD-L | Downstream model | CosyPose: Mask R-CNN detection stage plus a pose estimation stage |
| | Training recipe | Detection and pose stages trained on the same image type; strong data augmentation, identified by the source as a key component |
| | Training-set size | PBR: 350K pre-rendered BlenderProc4BOP images provided to participants. Render & paste v1: 50K images, 'obtained by OpenGL rendering of the 3D object models on random real photographs' |
| | Real evaluation split | BOP 2020 TUD-L test images |
| | Metric | AR (the source reports it in the ARCore table; the TUD-L column is the per-dataset AR) |
| | Source | Hodan et al., BOP Challenge 2020 (arXiv:2009.07378), Tab. 3 'The effect of different training images', TUD-L column |
| RarePlanes | Downstream model | Mask R-CNN |
| | Training recipe | Trained on the synthetic dataset only, three-class 'role' experiment (small, medium, large civil transport aircraft); each reported value is the average of 5 runs (39.46, sd 3.20) |
| | Training-set size | 50,000 synthetic satellite images with about 630,000 aircraft annotations |
| | Real evaluation split | Real RarePlanes test split |
| | Metric | mAP (IoU@[0.5:0.95]) |
| | Source | Shermeyer et al., RarePlanes: Synthetic Data Takes Flight, WACV 2021, Tab. 2, Mask R-CNN block, row 'role / Synth.' |
| ASD | Downstream model | SegFormer with the MiT-B0 backbone (HuggingFace Transformers) |
| | Training recipe | Trained on synthetic images only, i.e. the 12k : 0 synthetic-to-real ratio |
| | Training-set size | 12,000 synthetic images |
| | Real evaluation split | Hold-out set of 3,981 real soybean-field images; the table caption states all rows are tested on it |
| | Metric | Crop-class IoU from the source's mean-IoU table (weed and crop are reported separately; our score is the crop column) |
| | Source | Cieslak et al., Generating Diverse Agricultural Data for Vision-Based Farming Applications, 2024 (arXiv:2403.18351), Tab. 1, row '12k : 0', SegFormer columns |
| Polyp-Kvasir | Downstream model | UNet++ (26.1M parameters); the source also reports FPN and DeepLabv3+ |
| | Training recipe | The segmentation models and hyperparameters of the LDM baseline, 'with the same hyperparameter configurations except for altering the number of epochs to 100'; images resized to 256x256 |
| | Training-set size | 900 synthetic images, generated from the masks of the Kvasir-SEG training split |
| | Real evaluation split | 100 test images of Kvasir-SEG |
| | Metric | IoU |
| | Source | Dorjsembe et al., Polyp-DDPM, EMBC 2024 (arXiv:2402.04031), Tab. II, S900 block, UNet++ IoU, 'Test set-1' |
| Polyp-ETIS | Downstream model | UNet++ (26.1M parameters); the source also reports FPN and DeepLabv3+ |
| | Training recipe | The segmentation models and hyperparameters of the LDM baseline, 'with the same hyperparameter configurations except for altering the number of epochs to 100'; images resized to 256x256 |
| | Training-set size | 900 synthetic images, generated from the masks of the Kvasir-SEG training split |
| | Real evaluation split | 196 images of ETIS-LaribPolypDB |
| | Metric | IoU |
| | Source | Dorjsembe et al., Polyp-DDPM, EMBC 2024 (arXiv:2402.04031), Tab. II, S900 block, UNet++ IoU, 'Test set-3' |

Table 6. Per-variant provenance for all 21 rows. *Direction* is the direction in which the source benchmark evaluates transfer: every family trains on the synthetic source and tests on the real target, except Virtual KITTI 2, where the source evaluates a model trained on real imagery against the synthetic variants (see App. E). *Correspondence* states whether a row is genuinely aligned – the real and synthetic image are the same scene, matched by filename – or retrieval-based, where each real query is compared against $k$ sampled synthetic candidates. $n_q$ is the number of real queries; aligned rows have no candidate pool, and every retrieval row uses candidate seed 42. Per-family model, recipe, training-set size and evaluation split are in Table 5.

| Family | Variant | Real target | Synthetic source | $y$ | Metric | Direction | Source | Correspondence | $n_q$ | $k$ (seed) |
|---|---|---|---|---|---|---|---|---|---|---|
| ASD | cotton | ASD real field imagery | graphics + domain adaptation | 59.70 | mIoU | syn→real | Cieslak et al., 2024 | mode_b | 1,000 | 10 (42) |
| ASD | domain_adapted | ASD real field imagery | graphics + domain adaptation | 44.61 | mIoU | syn→real | Cieslak et al., 2024 | mode_b | 1,000 | 10 (42) |
| DIMO | randpose_randlight | DIMO real subset | graphics (BlenderProc) | 69.40 | COCO AP | syn→real | Vanherle et al., BMVC 2022 | mode_b | 1,000 | 10 (42) |
| DIMO | randpose_reallight | DIMO real subset | graphics (BlenderProc) | 65.80 | COCO AP | syn→real | Vanherle et al., BMVC 2022 | mode_b | 1,000 | 10 (42) |
| DIMO | realpose_randlight | DIMO real subset | graphics (BlenderProc) | 75.40 | COCO AP | syn→real | Vanherle et al., BMVC 2022 | mode_b | 1,000 | 10 (42) |
| DIMO | realpose_reallight | DIMO real subset | graphics (BlenderProc) | 78.50 | COCO AP | syn→real | Vanherle et al., BMVC 2022 | mode_a | 1,000 | – |
| Polyp-ETIS | ldm | ETIS-LaribPolypDB | generative | 36.64 | IoU | syn→real | Dorjsembe et al., EMBC 2024, Tab. II | mode_b | 196 | 10 (42) |
| Polyp-ETIS | polyp-ddpm | ETIS-LaribPolypDB | generative | 52.95 | IoU | syn→real | Dorjsembe et al., EMBC 2024, Tab. II | mode_b | 196 | 10 (42) |
| Polyp-ETIS | singan-seg | ETIS-LaribPolypDB | generative | 32.09 | IoU | syn→real | Dorjsembe et al., EMBC 2024, Tab. II | mode_b | 196 | 10 (42) |
| Polyp-Kvasir | ldm | Kvasir-SEG | generative | 66.94 | IoU | syn→real | Dorjsembe et al., EMBC 2024, Tab. II | mode_b | 100 | 10 (42) |
| Polyp-Kvasir | polyp-ddpm | Kvasir-SEG | generative | 71.56 | IoU | syn→real | Dorjsembe et al., EMBC 2024, Tab. II | mode_b | 100 | 10 (42) |
| Polyp-Kvasir | singan-seg | Kvasir-SEG | generative | 68.28 | IoU | syn→real | Dorjsembe et al., EMBC 2024, Tab. II | mode_b | 100 | 10 (42) |
| RarePlanes | test | RarePlanes real | graphics | 39.46 | COCO mAP | syn→real | Shermeyer et al., 2021 | mode_b | 1,000 | 10 (42) |
| TUD-L | train_bad_render_backgrounds | TUD-L real test | graphics (PBR / render-and-paste) | 49.50 | AR_Core | syn→real | BOP Challenge 2020, arXiv:2009.07378 | mode_b | 600 | 10 (42) |
| TUD-L | train_good_render | TUD-L real test | graphics (PBR / render-and-paste) | 68.50 | AR_Core | syn→real | BOP Challenge 2020, arXiv:2009.07378 | mode_b | 600 | 10 (42) |
| VKITTI | clone | KITTI / Cityscapes-trained model | graphics (Virtual KITTI 2) | 58.22 | mAP | real→syn | Cabon et al., arXiv:2001.10773, Tab. ss_rgb | mode_a | 1,000 | – |
| VKITTI | fog | KITTI / Cityscapes-trained model | graphics (Virtual KITTI 2) | 43.59 | mAP | real→syn | Cabon et al., arXiv:2001.10773, Tab. ss_rgb | mode_a | 1,000 | – |
| VKITTI | morning | KITTI / Cityscapes-trained model | graphics (Virtual KITTI 2) | 42.73 | mAP | real→syn | Cabon et al., arXiv:2001.10773, Tab. ss_rgb | mode_a | 1,000 | – |
| VKITTI | overcast | KITTI / Cityscapes-trained model | graphics (Virtual KITTI 2) | 58.31 | mAP | real→syn | Cabon et al., arXiv:2001.10773, Tab. ss_rgb | mode_a | 1,000 | – |
| VKITTI | rain | KITTI / Cityscapes-trained model | graphics (Virtual KITTI 2) | 46.14 | mAP | real→syn | Cabon et al., arXiv:2001.10773, Tab. ss_rgb | mode_a | 1,000 | – |
| VKITTI | sunset | KITTI / Cityscapes-trained model | graphics (Virtual KITTI 2) | 49.01 | mAP | real→syn | Cabon et al., arXiv:2001.10773, Tab. ss_rgb | mode_a | 1,000 | – |

Table 7. Dataset families used in the synthetic-data utility benchmark. The final correlation analysis is performed over dataset-level variants, not over individual image pairs. Pair-level metric computations stabilize the variant-level scores but do not increase the degrees of freedom of the final correlation test.

| Dataset family | Task / downstream metric | Pairing mode | Variants | Role in benchmark |
|---|---|---|---|---|
| DIMO [3] | Object detection; Mask R-CNN AP-style reported scores [18] | Mixed; aligned or retrieval depending on variant | 4 | Industrial metal objects; stresses lighting, pose, material reflectance, and geometric correspondence. |
| Virtual KITTI 2 [1] | Semantic segmentation; mAP as reported by the source (*reverse* transfer: trained on real, evaluated on the synthetic variants; see App. E) | Aligned | 6 | Driving scenes; stresses weather, illumination, and rendering-style variation while preserving scene structure. |
| TUD-L / BOP [7] | 6D pose estimation; $AR_{Core}$-style reported scores | Retrieval-based | 2 | Textureless object pose estimation; compares source-published rendering paradigms, with a confounded training budget (approximately 350K PBR versus 50K render-and-paste images). |
| RarePlanes [14] | Object detection; COCO mAP-style reported scores | Retrieval-based | 1 | Aerial synthetic-to-real transfer for aircraft detection. |
| ASD / SynSoy [2] | Crop/weed segmentation; the crop-class IoU as reported by the source | Retrieval-based | 2 | Agricultural synthetic-to-real transfer for crop/weed segmentation. |
| Polyp-Kvasir [5, 8] | Polyp segmentation; IoU-style reported scores | Retrieval-based | 3 | Colonoscopy imagery; three generative synthetic sources against the Kvasir-SEG target. |
| Polyp-ETIS [5, 15] | Polyp segmentation; IoU-style reported scores | Retrieval-based | 3 | Colonoscopy imagery; the same three generative sources against the harder ETIS-LaribPolypDB target. |

Table 8. The benchmark restricted to strictly synthetic-to-real transfer, i.e. excluding the six Virtual KITTI 2 variants. Pooled Pearson $r$ against the downstream reference. "Released calibration" freezes the coefficients and normalization fitted on all 21 variants; "refit" re-estimates both on the 15 remaining rows.

| Score | all 21 variants | 15 synthetic-to-real variants |
|---|---|---|
| SADGE, released calibration | $+0.834$ | $+0.841$ |
| SADGE, refit on the subset | – | $+0.857$ |
| SSIM | $+0.321$ | $+0.228$ |
| SP+LG | $+0.417$ | $+0.526$ |
| MASt3R | $+0.531$ | $+0.559$ |
| LoFTR | $+0.545$ | $+0.633$ |
| FID | $-0.008$ | $+0.017$ |
| DINOv2 | $+0.187$ | $+0.086$ |
| DINOv3 | $+0.232$ | $+0.054$ |
| SigLIP | $+0.129$ | $+0.006$ |
| SAM3 | $+0.244$ | $+0.218$ |
| PSNR | $+0.484$ | $+0.449$ |
| CLIP | $+0.164$ | $-0.057$ |
| LPIPS | $+0.585$ | $+0.537$ |

Table 9. Between- and within-family decomposition under both centering conventions. The family-mean columns reproduce the values reported in Table 2 of the main paper; the family-median columns report the alternative centering convention. RarePlanes contributes one variant and drops out of the within-family estimate either way, leaving 20 centered rows over six families.

| | family mean (reported) | | family median | |
|---|---|---|---|---|
| Score | between | within | between | within |
| SADGE | $+0.893$ | $+0.524$ | $+0.904$ | $+0.509$ |
| LPIPS | $+0.562$ | $+0.351$ | $+0.519$ | $+0.305$ |
| log LoFTR | $+0.677$ | $+0.235$ | $+0.614$ | $+0.204$ |
| log MASt3R | $+0.605$ | $+0.180$ | $+0.587$ | $+0.184$ |

Table 10. Six readings of the benchmark. Best score in each row in bold. All rows use the component convention – correspondence counts enter as $\log(1 + \text{inliers})$ – so the matcher columns differ from the raw-orientation values in Table 8. Rows two to five exclude RarePlanes, which contributes a single variant.

| Criterion | Unit | SADGE | LPIPS | log LoFTR | log MASt3R |
|---|---|---|---|---|---|
| Pooled Pearson $r$ (descriptive, in-sample) | 21 variants | $\mathbf{+0.834}$ | $+0.585$ | $+0.529$ | $+0.457$ |
| Within family, family-mean centred | 20 rows | $\mathbf{+0.524}$ | $+0.351$ | $+0.235$ | $+0.180$ |
| Within family, family-median centred | 20 rows | $\mathbf{+0.509}$ | $+0.305$ | $+0.204$ | $+0.184$ |
| Family-wise $z$-standardised, pooled | 20 rows, 6 families | $\mathbf{+0.586}$ | $+0.177$ | $+0.079$ | $+0.392$ |
| Family-balanced macro average of $r$ | 6 families | $\mathbf{+0.680}$ | $-0.021$ | $-0.234$ | $+0.071$ |
| Synthetic-to-real only (no VKITTI) | 15 variants | $\mathbf{+0.841}$ | $+0.537$ | $+0.686$ | $+0.721$ |

Table 11. Seed-induced spread against between-variant spread, on the transformed scale the analysis uses (correspondence counts as $\log(1 + \text{inliers})$). *seed s.d.* is the standard deviation over the five pools, averaged over the three variants; *between-variant s.d.* is the standard deviation of the five-pool means across those variants. A ratio near one means the pool draw is as large as the difference the proxy is being asked to detect.

| Proxy | Family | seed s.d. | between-variant s.d. | ratio |
|---|---|---|---|---|
| SSIM | Kvasir | 0.0043 | 0.0506 | 0.09 |
| SSIM | ETIS | 0.0057 | 0.0452 | 0.13 |
| PSNR | Kvasir | 0.0872 | 0.3720 | 0.23 |
| PSNR | ETIS | 0.0500 | 0.4543 | 0.11 |
| LPIPS | Kvasir | 0.0019 | 0.0075 | 0.26 |
| LPIPS | ETIS | 0.0016 | 0.0210 | 0.07 |
| FID | Kvasir | 0.5942 | 6.0855 | 0.10 |
| FID | ETIS | 0.2354 | 30.0944 | 0.01 |
| DINOv2 | Kvasir | 0.0023 | 0.0175 | 0.13 |
| DINOv2 | ETIS | 0.0018 | 0.0159 | 0.11 |
| DINOv3 | Kvasir | 0.0027 | 0.0204 | 0.13 |
| DINOv3 | ETIS | 0.0023 | 0.0189 | 0.12 |
| SP+LG | Kvasir | 0.0315 | 0.1567 | 0.20 |
| SP+LG | ETIS | 0.0221 | 0.0823 | 0.27 |
| LoFTR | Kvasir | 0.0419 | 0.1075 | 0.39 |
| LoFTR | ETIS | 0.0075 | 0.0865 | 0.09 |
| MASt3R | Kvasir | 0.1577 | 0.2806 | 0.56 |
| MASt3R | ETIS | 0.0663 | 0.1650 | 0.40 |

Table 12. Pooled correlation and induced rank of each re-drawn proxy over the 25 crossed pool assignments. The rank is identical in all 25 tables for every proxy.

| Proxy | pooled $r$ (seed 42) | min | max | range | rank | rank range |
|---|---|---|---|---|---|---|
| SSIM | $+0.3210$ | $+0.3100$ | $+0.3267$ | 0.0166 | 6 | 6 |
| PSNR | $+0.4840$ | $+0.4780$ | $+0.4862$ | 0.0082 | 3 | 3 |
| LPIPS | $+0.5854$ | $+0.5830$ | $+0.5878$ | 0.0048 | 1 | 1 |
| FID | $-0.0081$ | $-0.0108$ | $-0.0076$ | 0.0031 | 9 | 9 |
| DINOv2 | $+0.1873$ | $+0.1846$ | $+0.1924$ | 0.0077 | 8 | 8 |
| DINOv3 | $+0.2325$ | $+0.2313$ | $+0.2376$ | 0.0063 | 7 | 7 |
| SP+LG | $+0.3489$ | $+0.3417$ | $+0.3520$ | 0.0103 | 5 | 5 |
| LoFTR | $+0.5290$ | $+0.5280$ | $+0.5478$ | 0.0197 | 2 | 2 |
| MASt3R | $+0.4570$ | $+0.4525$ | $+0.4787$ | 0.0263 | 4 | 4 |

Table 13. The fitted metric and the family-disjoint held-out distribution over the same 25 assignments. Coefficients are reported as shares of $a + b + c$, because the fit is identified only up to positive scale.

| Quantity | seed 42 | min | median | max | range |
|---|---|---|---|---|---|
| pooled $r$, refitted | $+0.834$ | $+0.828$ | $+0.841$ | $+0.853$ | 0.025 |
| $a/(a+b+c)$ | $+0.282$ | $+0.280$ | $+0.284$ | $+0.288$ | 0.008 |
| $b/(a+b+c)$ | $+0.266$ | $+0.260$ | $+0.264$ | $+0.267$ | 0.007 |
| $c/(a+b+c)$ | $+0.452$ | $+0.451$ | $+0.452$ | $+0.454$ | 0.003 |
| family-disjoint median | $+0.729$ | $+0.714$ | $+0.742$ | $+0.762$ | 0.048 |
| family-disjoint worst | $-0.736$ | $-0.744$ | $-0.736$ | $-0.728$ | 0.016 |
| family-disjoint negative fraction | $+0.188$ | $+0.179$ | $+0.188$ | $+0.196$ | 0.018 |

Table 14. Bootstrap over variants (2,000 resamples of the 21 rows): pooled $r$ of the leading scores and of their differences, with 95% percentile intervals. This resamples rows independently and holds the fitted coefficients fixed, so it is descriptive only: it treats the 21 variants as exchangeable when they are clustered in families, and it therefore understates the true uncertainty.

| Quantity | point | bootstrap median | 95% interval |
|---|---|---|---|
| SADGE | $+0.834$ | $+0.840$ | [+0.731, +0.916] |
| LoFTR | $+0.545$ | $+0.551$ | [+0.172, +0.790] |
| LPIPS | $+0.585$ | $+0.602$ | [+0.184, +0.806] |
| log LoFTR | $+0.529$ | $+0.536$ | [+0.155, +0.779] |
| log MASt3R | $+0.457$ | $+0.460$ | [-0.023, +0.766] |
| SADGE $-$ LoFTR | $+0.289$ | $+0.283$ | [+0.093, +0.626] |
| SADGE $-$ log LoFTR | $+0.305$ | $+0.299$ | [+0.093, +0.658] |
| LoFTR $-$ LPIPS | $-0.040$ | $-0.035$ | [-0.363, +0.288] |

Table 15. Jackknife over variants: pooled $r$ after dropping one variant, for the three most and three least favourable deletions, and the range over all 21.

| Dropped variant | SADGE | LoFTR | log LoFTR |
|---|---|---|---|
| DIMO / realpose_reallight | 0.807 | 0.499 | 0.479 |
| DIMO / realpose_randlight | 0.823 | 0.464 | 0.460 |
| DIMO / randpose_randlight | 0.824 | 0.513 | 0.497 |
| VKITTI / overcast | 0.846 | 0.545 | 0.530 |
| Polyp-Kvasir / singan-seg | 0.853 | 0.590 | 0.563 |
| ASD / cotton | 0.855 | 0.572 | 0.566 |
| range over all 21 | 0.807–0.855 | 0.464–0.612 | 0.460–0.607 |

Table 16. Pooled $r$ of each correspondence matcher as a raw mean inlier count and after the $\log(1+\cdot)$ transform used inside the joint score.

| Matcher | raw mean inliers | $\log(1+\text{inliers})$ |
|---|---|---|
| MASt3R | 0.531 | 0.457 |
| SP+LG | 0.417 | 0.349 |
| LoFTR | 0.545 | 0.529 |

Table 17. Matched-capacity controls on the pooled criterion: the same $a\hat{G}+b\hat{A}+c\hat{G}\hat{A}$ form with $a, b, c \geq 0$ refitted to same-kind and cross-kind pairs. Direction is $(a, b, c)/(a+b+c)$; correspondence inputs are $\log(1+\text{inliers})$. These are in-sample and take a maximum over pairs, so they bound what the pooled criterion can show rather than establishing a held-out result.

| Pair | Kind | pooled $r$ | direction $(a, b, c)$ |
|---|---|---|---|
| DINOv3 × CLIP | appearance × appearance | 0.250 | (0.68, 0.00, 0.32) |
| DINOv2 × DINOv3 | appearance × appearance | 0.495 | (0.58, 0.00, 0.42) |
| LPIPS × DINOv2 | appearance × appearance | 0.641 | (0.69, 0.00, 0.31) |
| MASt3R × LoFTR | geometry × geometry | 0.558 | (0.10, 0.56, 0.34) |
| SP+LG × LoFTR | geometry × geometry | 0.530 | (0.00, 0.95, 0.05) |
| SP+LG × MASt3R | geometry × geometry | 0.463 | (0.00, 0.86, 0.14) |
| MASt3R × DINOv3 | geometry × appearance (released) | 0.834 | (0.28, 0.27, 0.45) |
| MASt3R × CLIP | geometry × appearance | 0.835 | (0.30, 0.27, 0.43) |
| LoFTR × LPIPS | geometry × appearance | 0.669 | (0.40, 0.45, 0.15) |

Table 18. Runtime benchmark for metric computation on `dimo_small` (1,000 pairs, CUDA), shown as a transposed table (metrics as columns). Load is one-time model initialization time. Total and Pairs/s report evaluation throughput for the full benchmark run.

| Metric | SSIM | PSNR | LPIPS | CLIP | SigLIP | SAM3 | DINOv2 | DINOv3 | MASt3R | SP+LG | LoFTR |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Load (s) | 0.02 | 0.01 | 1.07 | 3.82 | 3.01 | 6.16 | 7.38 | 1.55 | 7.10 | 0.86 | 0.29 |
| Total (s) | 8.96 | 6.13 | 15.42 | 32.65 | 88.27 | 581.56 | 133.64 | 31.52 | 271.46 | 38.00 | 47.93 |
| Pairs/s | 111.59 | 163.09 | 64.87 | 30.63 | 11.33 | 1.72 | 7.48 | 31.73 | 3.68 | 26.32 | 20.86 |

Table 19. Dataset licenses and access terms used in the benchmark. License terms should be checked against the official source before redistributing any raw or derived dataset files.

| Dataset | License / terms | Notes for SADGE use |
|---|---|---|
| DIMO | CC BY 4.0 | Public use is allowed with attribution and preservation of the license notice. No DIMO dataset files are redistributed by SADGE unless allowed under the license. |
| Virtual KITTI 2 | CC BY-NC-SA 3.0; non-commercial use only | Academic/non-commercial evaluation is permitted with attribution and ShareAlike requirements. Commercial use requires separate permission from the rights holder. |
| TUD-L / BOP | CC BY-SA 4.0 | Public use is allowed with attribution and ShareAlike requirements. TUD-L is used through the BOP benchmark distribution. |
| RarePlanes | CC BY-SA 4.0 | Public use is allowed with attribution and ShareAlike requirements. The dataset should be cited according to the RarePlanes instructions. |
| ASD / SynSoy | Permission-based / author-permission use | Used with permission from the dataset authors. We do not redistribute ASD/SynSoy files unless a public license or explicit redistribution permission is provided. If released later, the license and access terms will be included with the benchmark metadata. |
| Kvasir-SEG, ETIS-LaribPolypDB | Public research releases; per-provider terms | The two real colonoscopy targets are obtained from their original providers under the terms stated there; no images are redistributed. |
| LDM, SinGAN-Seg, Polyp-DDPM outputs | Per-source terms set by the original authors | The three generative synthetic sources are used as released by their authors; generated images are not redistributed. |

Table 20. External model and metric assets used for SADGE and baselines. The exact implementation and checkpoint source are recorded in the supplementary manifest.

| Asset | License / terms | Notes for SADGE use |
|---|---|---|
| DINOv2 [12] | Apache 2.0 | Used as an appearance encoder baseline/component. Copyright and license notices must be preserved if code or weights are redistributed. |
| DINOv3 [16] | DINOv3 License (Meta); gated on Hugging Face | Checkpoint `facebook/dinov3-vitl16-pretrain-lvd1689m` (ViT-L/16, LVD-1689M), revision `ea8dc286`. Used as the released SADGE appearance encoder. The DINOv3 License is not an open-source license: it restricts use, redistribution and derivatives, requires the license and notices to be passed on, and requires research publications that use DINOv3 or its outputs to acknowledge it. We use it for non-commercial academic research, do not redistribute the weights, and acknowledge DINOv3 here and in the citation above. |
| MASt3R [9] | Code: CC BY-NC-SA 4.0, non-commercial. Checkpoint: see the repository's `CHECKPOINTS_NOTICE` | Checkpoint `MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric`, from the `naver/mast3r` repository at commit `f5209afc`. Used as the released SADGE geometry matcher. The checkpoint carries its own terms in addition to the repository license: the `CHECKPOINTS_NOTICE` file lists the training datasets and their individual licenses, some of which are non-commercial or research-only, and those terms apply transitively to the released weights. Commercial use requires separate permission and independent verification against every dataset listed in that notice. We do not redistribute the weights. |
| LoFTR [17] | Apache 2.0 | Used as a geometry baseline/component. Copyright and license notices must be preserved. |
| LightGlue [10] | Apache 2.0 | Used as part of the SuperPoint+LightGlue baseline. The LightGlue repository releases its code and pretrained weights under Apache 2.0. |
| SuperPoint [4] | Restrictive license for the Magic Leap pretrained implementation/weights | If Magic Leap SuperPoint weights or inference files are used, their restrictive license must be respected. If an alternative implementation/checkpoint is used, the exact source and license should be specified in the manifest. |
| CLIP [13] | MIT for the OpenAI repository code | Used as an appearance baseline. The exact checkpoint/source is recorded in the manifest, and the MIT license notice is preserved where applicable. |
| SigLIP [20] | Apache-2.0 | Checkpoint `google/siglip-so400m-patch14-384`, revision `9fdffc58`. This is SigLIP v1: every `siglip_sim` value reported in this paper was computed with it, and no SigLIP 2 checkpoint was evaluated. |
| SAM3 [11] | SAM License (Meta); gated on Hugging Face. *Not* the Apache-2.0 terms of the original Segment Anything repository | Checkpoint `facebook/sam3`, revision `3c879f39`, used as an appearance baseline (vision-encoder last hidden state, mean-pooled). The evaluated model is SAM3, which is released under its own SAM License rather than under the Apache-2.0 license of the first Segment Anything release; the two must not be conflated. The SAM License restricts use and redistribution and carries an acceptable-use policy. We use it for non-commercial academic research and do not redistribute the weights. |
| LPIPS [21], PSNR, SSIM [19], FID [6] | Implementation-dependent | PSNR and SSIM are standard image metrics; LPIPS and FID depend on the implementation used. Package versions and licenses are recorded in the released environment manifest. |

## References


[1] Yohann Cabon, Naila Murray, and Martin Humenberger. Virtual kitti 2, 2020. 4, 10

[2] Mikolaj Cieslak, Umabharathi Govindarajan, Alejandro Garcia, Anuradha Chandrashekar, Torsten Hadrich, Aleksander Mendoza-Drosik, Dominik L. Michels, Soren Pirk, Chia-Chun Fu, and Wojciech Palubicki. Generating Diverse Agricultural Data for Vision-Based Farming Applications . In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 5422–5431, Los Alamitos, CA, USA, 2024. IEEE Computer Society. 10

[3] Peter De Roovere, Steven Moonen, Nick Michiels, and Francis Wyffels. Dataset of industrial metal objects. *arXiv preprint*, 2022. 10

[4] Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. Superpoint: Self-supervised interest point detection and description. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 224–236, 2018. 15

[5] Zolnamar Dorjsembe, Hsing-Kuo Pao, and Furen Xiao. Polyp-DDPM: Diffusion-based semantic polyp synthesis for enhanced segmentation. In *46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024. 10

[6] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In *Advances in Neural Information Processing Systems*, pages 6626–6637, 2017. 15

[7] Tomas Hodan, Frank Michel, Eric Brachmann, Wadim Kehl, Anders Glent Buch, Dirk Kraft, Bertram Drost, Joel Vidal, Stephan Ihrke, Xenophon Zabulis, Caner Sahin, Fabian Manhardt, Federico Tombari, Tae-Kyun Kim, Jiri Matas, and Carsten Rother. BOP: Benchmark for 6d object pose estimation. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018. 10

[8] Debesh Jha, Pia H. Smedsrud, Michael A. Riegler, Pål Halvorsen, Thomas de Lange, Dag Johansen, and Håvard D. Johansen. Kvasir-SEG: A segmented polyp dataset. In *International Conference on Multimedia Modeling (MMM)*, pages 451–462, 2020. 10

[9] Vincent Leroy, Yohann Cabon, and Jerome Revaud. MASt3R: Grounding image matching in 3d. *arXiv preprint arXiv:2406.09756*, 2024. 15

[10] Philipp Lindenberger, Paul-Edouard Sarlin, and Marc Pollefeys. Lightglue: Local feature matching at light speed. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 17627–17638, 2023. 15

[11] Meta AI. SAM 3: Segment anything with concepts. Model release, `https://huggingface.co/facebook/sam3`, 2025. Released under the SAM License. 15

[12] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc-Antoine Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaa El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Herve Jegou, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023. 15

[13] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 8748–8763. PMLR, 2021. 15

[14] Jacob Shermeyer, Thomas Hossler, Adam Van Etten, Daniel Hogan, Ryan Lewis, and Daeil Kim. Rareplanes: Synthetic data takes flight. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 207–217, 2021. 10

[15] Juan Silva, Aymeric Histace, Olivier Romain, Xavier Dhome, and Bertrand Granado. Toward embedded detection of polyps in WCE images for early diagnosis of colorectal cancer. *International Journal of Computer Assisted Radiology and Surgery*, 9(2):283–293, 2014. 10

[16] Oriane Siméoni, Huy Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025. 15

[17] Jiaming Sun, Zehong Shen, Yuang Wang, Hujun Bao, and Xiaowei Zhou. LoFTR: Detector-free local feature matching with transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8922–8931, 2021. 15

[18] Bram Vanherle, Steven Moonen, Frank Van Reeth, and Nick Michiels. Analysis of training object detection models with synthetic data. In *British Machine Vision Conference (BMVC)*, 2022. 10

[19] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4): 600–612, 2004. 15

[20] Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. SigLIP: Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11975–11986, 2023. 15

[21] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 586–595, 2018. 15